Zheyu Lin (林哲宇)

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Publication List

Education & Work Experience

- Postdoctoral researcher, Laboratoire AstroParticule et Cosmologie (APC), Paris, France 2025–2027
- Ph.D. in Astronomy, University of Science and Technology of China, Hefei, China 2020–2025
  Thesis: Investigating Tidal Disruption Events Based on Wide Field Sky Surveys
  Supervisors: Prof. Xu Kong（孔旭）& Prof. Ning Jiang（蒋凝）
- B.S. in Astronomy, University of Science and Technology of China, Hefei, China 2016–2020
  Thesis: Probing Tidal Disruption Events: Why, How?
  Supervisor: Prof. Xu Kong

Languages

Chinese - Mandarin (native), Teochew (native), Cantonese (fluent); English (fluent); French (learning)

Research Interests

My research interests tightly connect with transients around the supermassive black holes (SMBHs), especially tidal disruption events (TDEs) and turn-on active galactic nuclei (AGNs). My attention is mainly paid to these.

- Enlarge the sample of transients around the SMBHs by exploiting time-domain and spectroscopic survey data, and summarize their collective features.
- Unveil the nature of TDEs and turn-on AGNs by multiwavelength emission and spectroscopy.
- Explore the unknown parameter space, trying to find new kinds or features of transients.

Since my postdoc career at APC, I joined the Space-based multi-band astronomical Variable Objects Monitor (SVOM) and Einstein Probe (EP) collaboration as an affiliated scientist of Dr. Alexis Coleiro and started learning how to interpret the multiwavelength evolution of gamma-ray bursts and fast X-ray transients.

Projects

- Affiliated scientist of SVOM and EP 2025–Present
  - Follow-up and characterization of gamma-ray bursts and fast X-ray transients.
- Long-term monitoring of nuclear transients 2022–Present
  - Experience in Swift / XMM-Newton / EP / NICER / spectroscopy applications (>800 ks/>150 ks/>300 ks/30 ks/>20 nights awarded) and data analysis.
  - Expertise in collecting multiwavelength data and discovering meaningful features.
  - Two papers published in the ApJ Letters.

- Algorithms of searching TDEs 2024–Present
  - Development of AI-based TDE classification algorithms with Ph.D. candidate Ranfang Zheng, Prof. Xu Kong, et al.
  - Two papers published in ApJ as a corresponding author. One on the photometric classifier and the other on the spectroscopic classifier.
- Searching TDEs by the Wide Field Survey Telescope (WFST) 2020–2025
  - One mock observation paper published in the MNRAS.
  - Built the WFST differential photometry database with Prof. Jie Lin, et al.
  - Built useful parametric filters on WFST and discovered a few TDE candidates.

Scholarships

- CAS President Award - Excellence Prize, 2025.
- National Scholarship for Ph.D. Students, 2024.
- CASC Scholarship, First Prize, 2023.
- WFST Scholarship, 2022, 2023, 2024.

Positions of Responsibility

- Reviewer for The Astrophysical Journal (ApJ) and Research in Astronomy and Astrophysics (RAA)
- Volunteer for USTC Astronomy Summer Camp Summer 2022
- Teaching Assistant for ASTR6003P Fall 2021
- Minister of Academy for USTC English Club (EC) Fall 2017 - Summer 2018
  - The designer and organizer of "USTC Mystery Hunt" – an outdoor mystery solving game. Following the great success, it has been held as a (bi-)annual activity for seven times and imitated by other clubs
  - The creator of the award-winning official account of EC
- Assistant of Study Affairs for Class 2, Grade 2016, School of Physics Fall 2016 - Summer 2017

Press (in English)

- This unlucky star got mangled by a black hole – twice, *Nature Research Highlights* August 2024
- 'Unluckiest star' may be trapped in deadly dance with a black hole, *New Scientist* May 2024

Selected Talks

- X-ray Quasi-Periodic Eruptions and Repeating Nuclear Transients workshop, ESAC, Madrid June 2025
- Transient Phenomena and Physical Processes around SMBHs, Tsung-Dao Lee Institute October 2024
- TDEs and NTs: Entering the Data-Rich Era, Heraklion, Crete, Greece September 2024
- Einstein Probe/WFST time-domain science seminar, Lenghu, Qinghai August 2023

Selected Awards

- Outstanding PhD graduate of USTC & Anhui Province, 2025.
- One of the top ten progresses of the year, School of Physics, USTC, 2024.
- Second place in the 1st popular science article competition by the National Astronomical Observatories of the Chinese Academy of Sciences (NAOC), 2024. Awarded article: *Hunting the beasts in the center of galaxies: A story of supermassive black holes* (Chinese:《星系中心探“巨兽”——超大质量黑洞的故事》).
- “Gold dolphin” award of 50m breaststroke swimming (finish in 50 seconds), 2023.
- “Outstanding official account” award for the USTC English Club, as the creator and administrator, 2018.

Publications by Oct 22 2026

Full List on NASA/ADS

6 Papers as the First or Corresponding Author:

6. Zheng Ranfang, **Lin Zheyu**, Kong Xu, et al., Two Earliest Optical-UV Tidal Disruption Events Hidden in the SDSS DR7 Catalog Unveiled by the Transformer-Based Spectrum Classifier, ApJ, 1007, 140.
5. Zheng Ranfang, **Lin Zheyu**, Kong Xu, et al., TTC: Transformer-based TDE Classifier for the Wide Field Survey Telescope (WFST), 2026, ApJ, 999, 181. (Cited 4)
4. **Lin Zheyu**, Jiang Ning, Wang Yibo, Kong Xu, et al., Insights from the “Red devil” AT 2022fpx: A dust-reddened TDE family excluded by their apparent red color?, 2025, ApJ, 990, 22. (Cited 6)
3. **Lin Zheyu**, Jiang Ning, Wang Tinggui, Kong Xu, et al., The unluckiest star: A spectroscopically confirmed repeated partial tidal disruption event AT 2022dbl, 2024, ApJL, 971, L26. (Cited 47, selected as Nature research highlight)
2. **Lin Zheyu**, Jiang Ning, Kong Xu, et al., The Luminosity Function of Tidal Disruption Flares for the ZTF-I Survey, 2022, ApJL, 939, L33. (Cited 35)
1. **Lin Zheyu**, Jiang Ning & Kong Xu, The prospects of finding tidal disruption events with 2.5-m Wide-Field Survey Telescope based on mock observations, 2022, MNRAS, 513, 2422. (Cited 35)

19 Papers as the co-author:

19. Shifeng Huang, Wang Tinggui, Jiang Ning, et al., A Unified Timescale Relation for Quasi-Periodic Eruptions and Repeated Nuclear Transients, 2026, ApJL, 1009, 23.
18. Zhuang Xiaoyu, Ran Song, Meng Dezheng, et al., Searching for Type Ia Supernovae in the Dark Energy Spectroscopic Instrument, 2026, ApJ, 1004, 198.
17. Li Jiaxun, Wang Tinggui & **Lin Zheyu**, Characterizing and spectrally modeling embedded FUor eruptions in the near-infrared, 2026, A&A, 710, 83.
16. Huang Mengqiu, Xue Yongquan, Li Shuo, et al., The Faintest, Extremely Variable X-ray Tidal Disruption Event from a Supermassive Black Hole Binary?, 2026, The Innovation, 7, 3, 101169.

15. Huang Shifeng, Wang Tinggui, Jiang Ning, et al., Unveiling the Cosmic Dance of Repeated Nuclear Transient ASASSN-14ko: Insights from Multiwavelength Observations, 2025, ApJ, 988, 237.

14. Ding Weiyu, Kong Weiwen, Sui Jipeng, et al., Exploring the Link between Extreme Coronal Line Emitters and Tidal Disruption Events, 2025, ApJ, 988, 187.

13. Luo Yibin, Fan Lulu, Sun Weibin, et al., The $M_{\rm BH}$–$M_\star$ Relation of the Hyperluminous Dust-obscured Quasars up to $z \sim 4$, 2025, ApJ, 986, 195.

12. Lin Jie, Wang Tinggui, Cai Minxuan, et al., Minute-cadence observations on Galactic plane with Wide Field Survey Telescope (WFST): Overview, methodology and early results, 2025, ApJS, 278, 29.

11. Yao Yao, Wang Enci, He Zhicheng, **Lin Zheyu**, et al., Bipolar blobs as evidence of hidden AGN activities in the low-mass galaxies, 2024, ApJL, 972, L16.

10. Luo Yibin, Fan Lulu, Liang Yongming, et al., Lyα imaging around the hyperluminous dust-obscured quasar W2246−0526 at $z = 4.6$, 2024, ApJ, 972, 51.

9. Wang Yibo, Wang Tinggui, Jiang Ning, et al., ASASSN-18ap: A Dusty Tidal Disruption Event Candidate with an Early Bump in the Light Curve, 2024, ApJ, 966, 136.

8. Huang Shifeng, Jiang Ning, Zhu Jiazheng et al., AT 2023lli: A Tidal Disruption Event with Prominent Optical Early Bump and Delayed Episodic X-Ray Emission, 2024, ApJL, 964, L22.

7. Chen Lijun, Zhang Hong-Xin, Lin Zesen, et al., Dwarf Galaxies with the Highest Concentration Are Not Thicker than Ordinary Dwarf Galaxies, 2023, ApJ, 958, 117.

6. Huang Shifeng, Jiang Ning, **Lin Zheyu**, et al., AT2018dyk revisited: a tidal disruption event candidate with prominent infrared echo and delayed X-ray emission in a LINER galaxy, 2023, MNRAS, 525, 4057.

5. Wang Tinggui, Liu Guilin, Cai Zhenyi, et al., Science with the 2.5-meter Wide Field Survey Telescope (WFST), 2023, SCPMA, 66, 109512.

4. Zhu Jiazheng, Jiang Ning, Wang Tinggui, et al., AT 2023clx: The Faintest and Closest Optical Tidal Disruption Event Discovered in Nearby Star-forming Galaxy NGC 3799, 2023, ApJL, 952, L35.

3. Liu Zheng-Yan, **Lin Zhe-Yu**, Yu Ji-Ming, et al., Target-of-Opportunity Observation Detectability of Kilonovae with WFST, 2023, ApJ, 947, 59.

2. Luo Yibin, Fan Lulu, Zou Hu, et al., An Overdensity of Red Galaxies around the Hyperluminous Dust-obscured Quasar W1835+4355 at $z = 2.3$, 2022, ApJ, 935, 80.

1. Wang Yibo, Jiang Ning, Wang Tinggui, et al., Discovery of ATLAS17jrp as an Optical-, X-Ray-, and Infrared-bright Tidal Disruption Event in a Star-forming Galaxy, 2022, ApJL, 930, L4.